

CREDIT CARD FRAUD DETECTION USING MACHINE LEARNING

Project Report

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ABSTRACT

Credit card fraud is a major challenge in the financial sector, resulting in significant financial losses and affecting customer trust. This project focuses on developing a machine learning-based system for detecting fraudulent credit card transactions. The dataset contains transaction-related features, including Time, Amount, and anonymized variables (V1–V28), along with the Class variable representing legitimate and fraudulent transactions.

The project follows a systematic process involving data preprocessing, exploratory data analysis (EDA), feature scaling, handling of class imbalance, and machine learning model development. Logistic Regression with balanced class weights was implemented to improve the detection of fraudulent transactions. Model performance was evaluated using appropriate classification metrics, with particular attention to fraud detection capability. The model was also interpreted to identify the transaction features that have the greatest influence on fraud predictions.

In addition, an interactive Power BI dashboard was developed to visualize transaction patterns, fraud distribution, transaction amounts, fraud rates, and other important trends. The final system provides both predictive and visual analytical capabilities that can help identify potentially fraudulent transactions and support data-driven decision-making.

1. INTRODUCTION

Credit card fraud is one of the major challenges faced by the financial industry. With the increasing use of online transactions and digital payment systems, the number of electronic transactions has grown significantly. This has also created opportunities for fraudulent activities, making reliable fraud detection systems increasingly important.

Traditional rule-based fraud detection methods may not always be effective because fraudulent transaction patterns can change over time. Machine learning provides a data-driven approach that can learn patterns from historical transaction data and identify transactions that have characteristics similar to previously observed fraudulent transactions. This project focuses on developing a machine learning-based credit card fraud detection system using transaction data. The dataset contains numerical transaction features, including Time, Amount, and anonymized variables V1 to V28, along with a Class variable indicating whether a transaction is legitimate or fraudulent.

The project involves data preprocessing, exploratory data analysis (EDA), feature scaling, handling the imbalanced nature of the dataset, and developing a Logistic Regression classification model. The model is evaluated using classification metrics to understand its ability to distinguish fraudulent transactions from legitimate ones. In addition to the machine learning model, a Power BI dashboard is developed to provide an interactive view of transaction patterns, fraud distribution, transaction amounts, fraud rates, and other important trends. The combination of predictive modelling and visualization provides a practical approach to understanding and detecting credit card fraud.

2. OBJECTIVES

To analyze credit card transaction data and understand the characteristics and distribution of legitimate and fraudulent transactions.

To perform data preprocessing including data cleaning, feature scaling, and preparation of the dataset for machine learning.

To handle the class imbalance problem because fraudulent transactions represent only a small proportion of the overall transactions.

To develop a machine learning model using Logistic Regression to classify transactions as legitimate or fraudulent.

To evaluate model performance using suitable classification metrics and assess its ability to detect fraudulent transactions.

To interpret the model results and identify the top transaction features that contribute to fraud predictions.

To develop an interactive Power BI dashboard that presents fraud trends, transaction patterns, fraud rates, and high-risk segments.

To provide useful insights that can support better monitoring and identification of potentially fraudulent credit card transactions.

3. DATASET DESCRIPTION

The dataset used in this project is a credit card transaction dataset designed for fraud detection. It contains transaction-level information with numerical features that describe different characteristics of each transaction.

The dataset consists of 30 input features, including Time, V1 to V28, and Amount. The features V1 to V28 are anonymized numerical variables. The Class variable is the target variable used to identify whether a transaction is legitimate or fraudulent.

The target variable Class contains two categories:

- **0** – Legitimate transaction
- **1** – Fraudulent transaction

The dataset is highly imbalanced because fraudulent transactions form only a small proportion of the total transactions. Therefore, class imbalance was considered during the model development process. The data was prepared for machine learning through preprocessing and feature scaling before training the classification model.

The dataset was divided into training, validation, and testing subsets. The training data was used to build the Logistic Regression model, while the validation data was used for model evaluation and threshold-related analysis. The final testing data was kept for assessing the model's performance on unseen transactions.

```
[27]: # Display the first 5 rows
df.head()
```

	Time	V1	V2	V3	V4	V5	V6	V7	V8	V9	...	V21	V22	V23	V24	V26	V27	V28	Amount	Class
0	0.0	-1.359807	-0.072781	2.536347	1.378155	-0.338321	0.462388	0.239599	0.098698	0.363787	...	-0.018307	0.277838	-0.110474	0.066928	0.128539	-0.189115	-0.021053	149.62	0
1	0.0	1.191857	0.266151	0.166480	0.448154	0.060018	-0.082361	-0.078803	0.085102	-0.255425	...	-0.225775	-0.638672	0.101288	-0.339846	0.167170	0.125895	-0.008983	2.69	0
2	1.0	-1.358354	-1.340163	1.773209	0.379780	-0.503198	1.800499	0.791461	0.247676	-1.514654	...	0.247998	0.771679	0.909412	-0.689281	-0.327642	-0.139097	-0.059752	378.66	0
3	1.0	-0.966272	-0.185226	1.792993	-0.863291	-0.010309	1.247203	0.237609	0.377436	-1.387024	...	-0.108300	0.005274	-0.190321	-1.175575	0.647376	-0.221929	0.062723	123.50	0
4	2.0	-1.158233	0.877737	1.548718	0.403034	-0.407193	0.095921	0.592941	-0.270533	0.817739	...	-0.009431	0.798278	-0.137458	0.141267	-0.206010	0.502292	0.215153	69.99	0

5 rows × 31 columns

Figure 3.1 Dataset Preview – First Five Rows

4. METHODOLOGY

The project follows a systematic machine learning workflow for detecting fraudulent credit card transactions. The methodology consists of data preparation, exploratory data analysis, preprocessing, model development, model evaluation, interpretation, and visualization.

4.1 Data Preprocessing

The transaction dataset was examined to understand its structure, features, data types, and target variable. The input features were separated from the target variable, where Class represents whether a transaction is legitimate or fraudulent. The numerical features were prepared for machine learning, and feature scaling was performed to bring the variables to a comparable scale.

	Time	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14	V15	V16	V17	V18	V19	V20	V21	V21	V22	V24	V25	V26	V27	V28	V28	Amount	Cu	
count	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	284807.0	
mean	94813.859575	1.799000e-12	4.251200e-13	3.653240e-13	8.321470e-13	1.608360e-13	4.248460e-13	3.054630e-13	8.779460e-14	-1.179730e-12	1.403120e-12	6.577000e-12	-8.624720e-12	8.577030e-13	1.187702e-12	2.113960e-12	1.637180e-12	1.805450e-12	1.885450e-12	6.837520e-13	1.285310e-12	5.487720e-13	3.483730e-13	5.732820e-13	-3.845350e-13	9.037200e-13	6.987000e-13	5.877000e-13	8.134610e-13	88.349619	0.001727		
std	47488.145955	1.958000e+00	1.651300e+00	1.516255e+00	1.415860e+00	1.380240e+00	1.332270e+00	1.237890e+00	1.194330e+00	1.098630e+00	1.088050e+00	1.030710e+00	1.030710e+00	9.820130e+01	9.590140e+01	9.590120e+01	9.510120e+01	9.538230e+01	6.780060e+01	6.671940e+01	8.389320e+01	8.143340e+01	8.143360e+01	7.653260e+01	7.216890e+01	3.145240e+01	2.578100e+01	3.122780e+01	4.622770e+01	5.880330e+01	258.120109	0.041527	
min	0.000000	-5.640750e+01	-7.271570e+01	-4.832590e+01	-5.683710e+01	-1.137430e+02	-2.618810e+01	-4.285720e+01	-7.321670e+01	-1.344870e+01	-2.940160e+01	-4.701020e+01	-8.705470e+01	-8.705470e+01	-8.705470e+01	-8.705470e+01	-8.705470e+01	-8.705470e+01	-8.705470e+01	-8.705470e+01	-8.705470e+01	-8.705470e+01	-8.705470e+01	-8.705470e+01	-8.705470e+01	-8.705470e+01	-8.705470e+01	-8.705470e+01	-8.705470e+01	-8.705470e+01	-8.705470e+01	0.000000	0.000000
25%	54201.500000	-9.203730e+01	-5.985490e+01	-4.903640e+01	-8.486410e+01	-4.913970e+01	-7.682950e+01	-5.540750e+01	-2.086290e+01	-6.430970e+01	-5.597320e+01	-7.626300e+01	-4.835380e+01	-4.835380e+01	-4.835380e+01	-4.835380e+01	-4.835380e+01	-4.835380e+01	-4.835380e+01	-4.835380e+01	-4.835380e+01	-4.835380e+01	-4.835380e+01	-4.835380e+01	-4.835380e+01	-4.835380e+01	-4.835380e+01	-4.835380e+01	-4.835380e+01	-4.835380e+01	-4.835380e+01	5.600000	0.000000
50%	84692.000000	1.810000e+02	6.548550e+02	1.794430e+01	-1.984630e+02	-5.433030e+02	-2.710170e+01	4.010300e+02	2.235000e+02	-5.142970e+02	-9.020910e+02	-3.972910e+02	1.380150e+02	1.380150e+02	1.570260e+02	1.570260e+02	1.566640e+02	6.657460e+02	4.903240e+02	-6.685040e+02	1.663540e+02	-6.076140e+02	4.261130e+02	7.342460e+02	2.770100e+02	6.244030e+02	6.054710e+02	4.622170e+02	4.693310e+02	3.300930e+02	22.000000	0.000000	
75%	139538.500000	1.315620e+03	8.037290e+02	1.027190e+03	7.431410e+03	6.115040e+03	3.905640e+03	5.704300e+03	3.273430e+03	5.971390e+03	4.789520e+03	1.383270e+04	4.388990e+03	6.629430e+03	4.933360e+03	4.933360e+03	6.433330e+03	6.433330e+03	5.253930e+03	4.644010e+03	5.487730e+03	4.598570e+03	1.661590e+03	1.363770e+03	5.253360e+03	1.974240e+03	1.307510e+03	2.400520e+03	9.915120e+02	9.190120e+02	77.365000	0.000000	
max	172792.000000	2.454030e+03	2.205730e+03	5.382550e+03	1.687530e+04	1.4803470e+04	7.330030e+03	1.203890e+04	2.000720e+04	1.519490e+04	1.8877870e+04	1.275040e+04	7.884020e+03	7.884020e+03	7.884020e+03	7.884020e+03	7.884020e+03	7.884020e+03	7.884020e+03	7.884020e+03	7.884020e+03	7.884020e+03	7.884020e+03	7.884020e+03	7.884020e+03	7.884020e+03	7.884020e+03	7.884020e+03	7.884020e+03	7.884020e+03	2581.160000	1.000000	

8 rows × 31 columns

Dataset Preview – Descriptive Statistics (All Features)

Figure 4.1.1 Dataset Processing Preview

4.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand transaction distributions and identify patterns associated with fraudulent transactions. The analysis included examining the class distribution, transaction amounts, transaction timing, and relationships between transaction features and the fraud class.

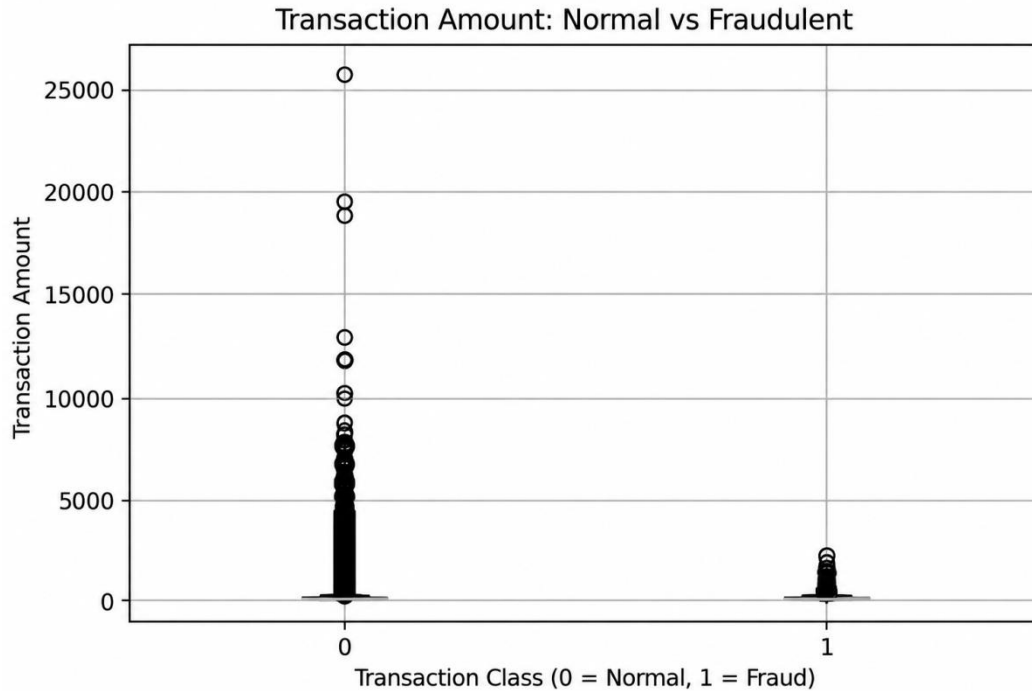


Figure 4.2.1 Transaction Amount in EDA

4.3 Handling Class Imbalance

The dataset contains significantly fewer fraudulent transactions than legitimate transactions. To address this class imbalance during model training, balanced class weights were used in the Logistic Regression model. This gives greater importance to the minority fraud class and helps the model pay more attention to fraudulent transactions.

4.4 Model Development

A Logistic Regression classification model was selected for fraud detection. The model was trained using the prepared and scaled training data. The model estimates the probability that a transaction belongs to the fraudulent class and uses this probability to classify transactions.

4.5 Model Evaluation

The model was evaluated using classification performance measures, with particular attention to its ability to identify fraudulent transactions. Validation data was used to assess the model and analyze fraud probabilities before evaluating the final model on unseen test data.

4.6 Model Interpretation

The coefficients generated by the Logistic Regression model were analyzed to identify the features that have the greatest influence on fraud predictions. The absolute coefficient values were used to rank the top predictors of fraud.

4.7 Power BI Visualization

A Power BI dashboard was created to present important findings from the transaction dataset. The dashboard includes visualizations of transaction counts, fraud rate, transaction amounts, fraud trends over time, and comparisons between legitimate and fraudulent transactions.

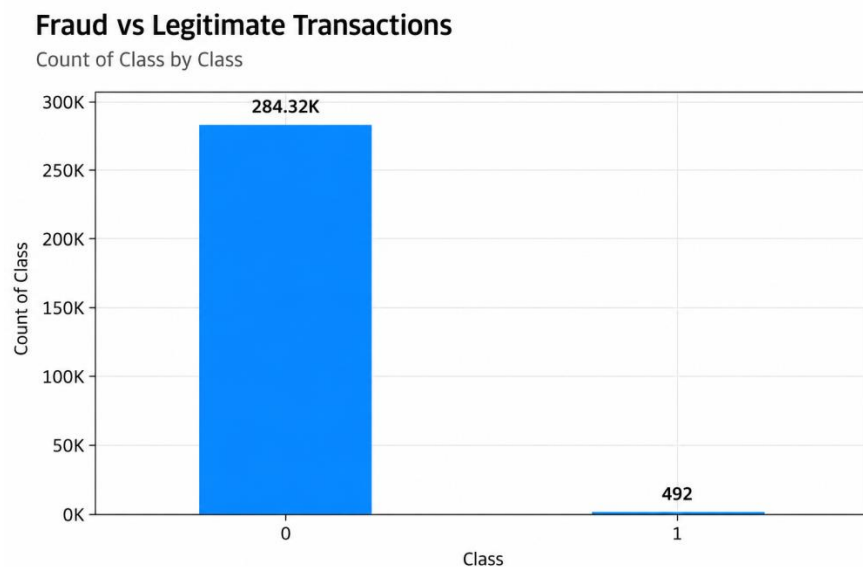


Figure 4.7.1 Fraud vs Legitimate Transactions

5. EDA FINDINGS

Exploratory Data Analysis (EDA) was performed to understand the structure of the credit card transaction dataset and identify patterns associated with fraudulent transactions. The analysis focused on the target-class distribution, transaction amounts, transaction timing, and differences between legitimate and fraudulent transactions.

5.1 Class Distribution

The dataset contains 284,807 transactions. Among these, 284,315 transactions (99.8273%) are legitimate, while only 492 transactions (**0.1727%**) are fraudulent. This demonstrates a severe class imbalance, with fraudulent transactions representing a very small proportion of all transactions. This imbalance is an important consideration when developing and evaluating the fraud detection model.

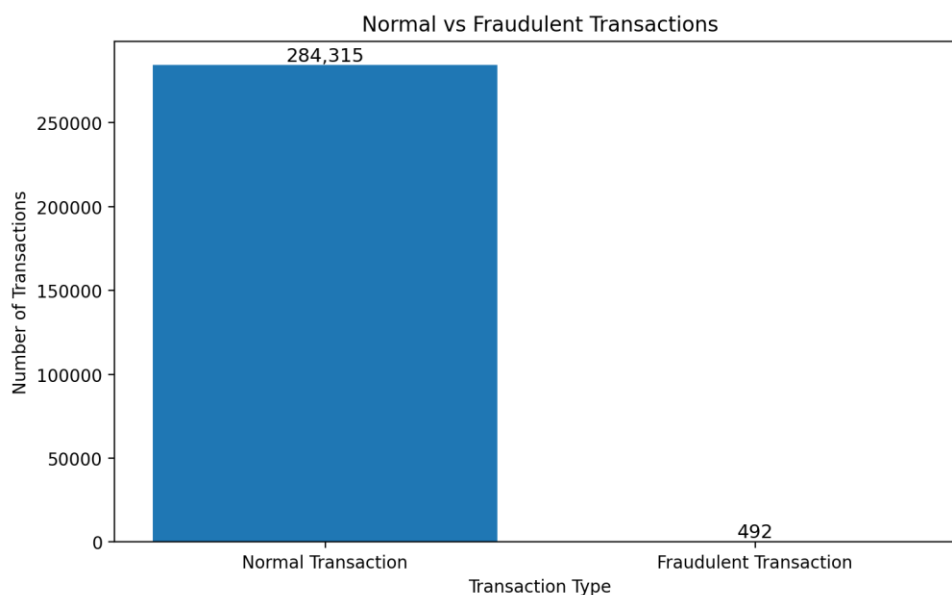


Figure 5.1.1 Normal vs Fraudulent Transactions

5.2 Data Quality

The initial data inspection showed that the dataset did not contain missing values. During data cleaning, 1,081 duplicate records were identified and removed. After removing the duplicates, the dataset contained 283,726 unique transactions. This preprocessing step helped ensure that repeated records did not unnecessarily influence the analysis and model.

5.3 Transaction Amount

Transaction amount was analyzed to understand whether the monetary value of transactions differed between legitimate and fraudulent transactions. The analysis showed that fraudulent transactions had a higher average transaction amount than legitimate transactions in the analyzed data. In the Power BI dashboard, the average transaction amount was approximately \$120 for fraudulent transactions compared with approximately \$88 for legitimate transactions.

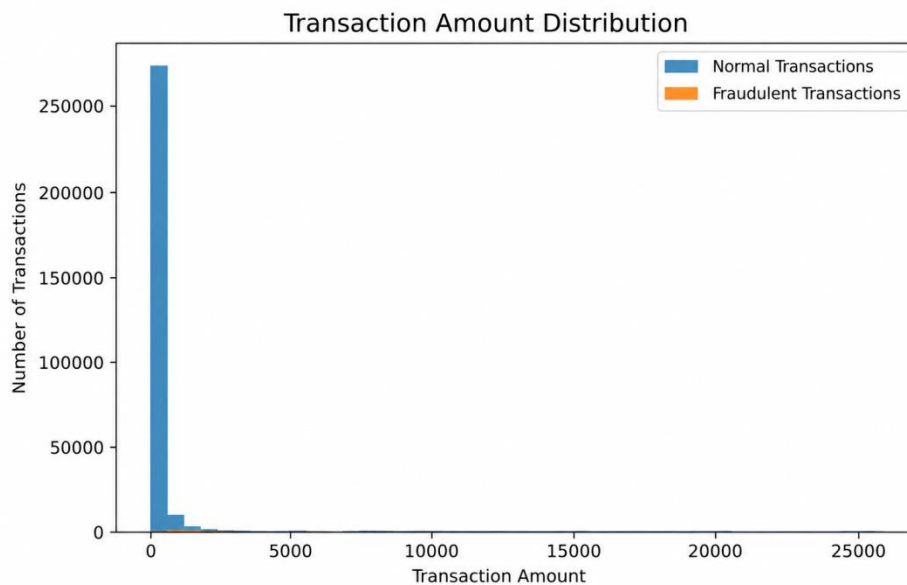


Figure 5.3.1 Transaction Amount Distribution

5.4 Transaction Time

Transaction timing was also examined to identify changes in transaction activity and fraud patterns across time. The time-based analysis provides a way to observe periods where fraudulent activity may differ from normal transaction activity and supports further investigation of potentially high-risk periods.

4. Top 10 Features Differing Between Normal and Fraudulent Transactions

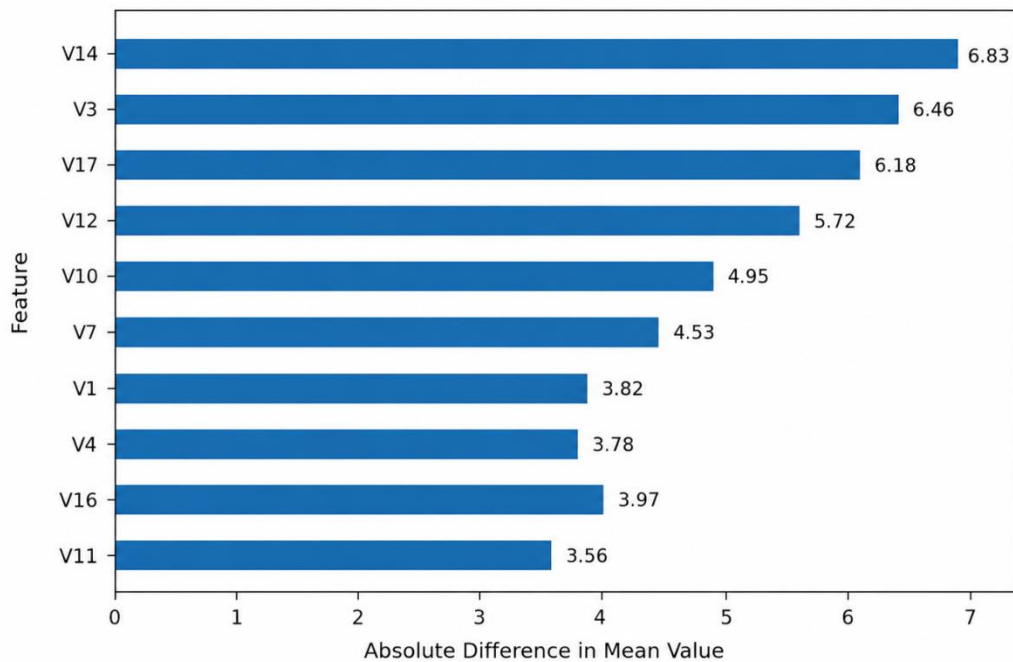


Figure 5.4.1 Top 10 Features Differing Between Normal and Fraudulent Transactions

5.5 Key EDA Insights

The EDA identified three important characteristics of the dataset. First, fraudulent transactions form a very small minority of all transactions, creating a significant class imbalance. Second, transaction amount shows a difference between fraudulent and legitimate transactions. Third, transaction timing can be used to explore patterns in fraud activity. These findings guided the subsequent preprocessing, model development, and dashboard design.

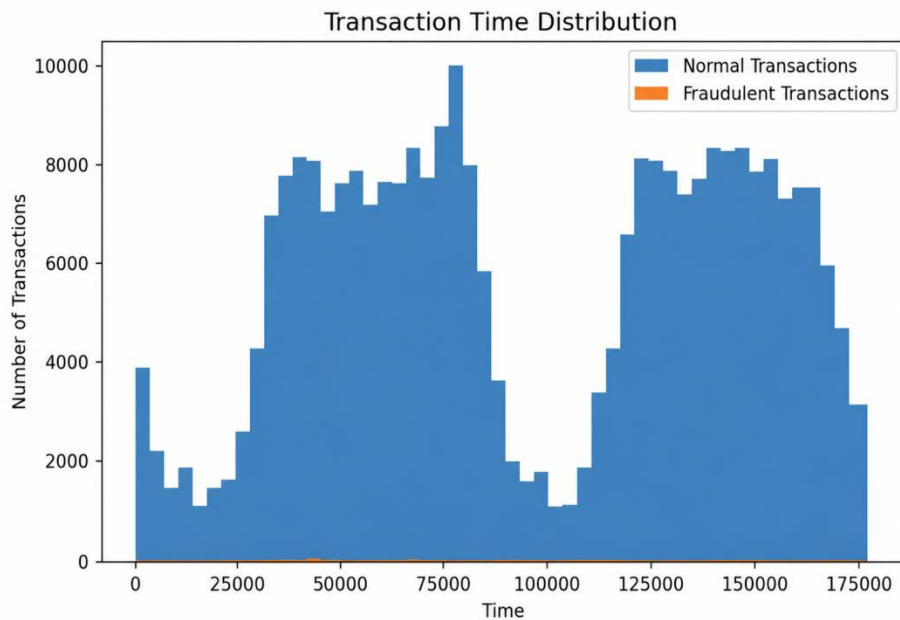


Figure 5.5.1 Transaction Time Distribution of Normal and Fraudulent Transactions

6. MODEL DEVELOPMENT & PERFORMANCE

6.1 Model Selection

Logistic Regression was selected as the classification algorithm for this project. It is a supervised learning algorithm suitable for binary classification problems such as distinguishing between legitimate and fraudulent transactions. It also provides interpretable coefficients, which can be used to identify the features that have the greatest influence on fraud predictions.

Top 10 Predictors of Fraud:

	Feature	Coefficient	Importance
29	Amount	2.311589	2.311589
1	V1	1.652232	1.652232
14	V14	-1.440515	1.440515
4	V4	1.279250	1.279250
10	V10	-1.265119	1.265119
12	V12	-1.176368	1.176368
5	V5	1.120480	1.120480
2	V2	0.994081	0.994081
17	V17	-0.953312	0.953312
20	V20	-0.940625	0.940625

Figure 6.1.1 Logistic Regression of Top 10 Predictors of Fraud

6.2 Feature Scaling

Feature scaling was performed using StandardScaler. Scaling transforms the numerical input variables to a comparable scale, which is particularly important for Logistic Regression because the model's coefficients are influenced by the scale of the input features.

```
Scaled training shape: (182276, 30)
Scaled validation shape: (45569, 30)
Scaled testing shape: (56962, 30)
```

Figure 6.2.1 The StandardScaler of Logistic Regression

6.3 Handling Class Imbalance

The dataset is highly imbalanced, with fraudulent transactions representing only a small percentage of the total transactions. To reduce the effect of this imbalance during training, the Logistic Regression model was configured with `class_weight='balanced'`. This assigns greater weight to the minority fraud class and helps the model identify fraudulent transactions more effectively.



Figure 6.3.1 Class weight Balancing with Fraud vs Legitimate

6.4 Model Training

The preprocessed dataset was divided into training and testing data. The Logistic Regression model was trained using the scaled training features and the corresponding class labels. After training, the model generated fraud probabilities and class predictions for previously unseen transactions.

```
Training set: (227845, 30)
Testing set: (56962, 30)
Training fraud cases: 394
Testing fraud cases: 98
```

Figure 6.4.1 The Training and Testing of Preporcessed Dataset

6.5 Model Performance

The model achieved a ROC-AUC score of 0.9737, approximately 0.974, on the test data. ROC-AUC measures the model's ability to distinguish between fraudulent and legitimate transactions across different classification thresholds. A value close to 1 indicates strong discriminatory performance.

	Predicted Legitimate	Predicted Fraud
Actual Legitimate	56,842	22
Actual Fraud	18	80

Table 6.5.1 The confusion matrix obtained from the test data

The model correctly identified 80 fraudulent transactions and incorrectly classified 18 fraudulent transactions as legitimate. It also incorrectly classified 22 legitimate transactions as fraudulent. These results demonstrate that the model was able to detect a substantial proportion of fraudulent transactions while keeping false positives relatively low.

```
Confusion Matrix:
[[56842  22]
 [  18  80]]

Classification Report:
              precision    recall  f1-score   support

     0       1.00      1.00      1.00     56864
     1       0.78      0.82      0.80        98

 accuracy          1.00     56962
 macro avg       0.89      0.91      0.90     56962
 weighted avg    1.00      1.00      1.00     56962
```

Figure 6.5.1 The Confusion Matrix and Classification Report

6.6 Model Evaluation Summary

Overall, the Logistic Regression model demonstrated strong ability to separate fraudulent transactions from legitimate transactions, as reflected by the ROC-AUC score of 0.9737. The use of balanced class weights was important because of the severe class imbalance in the dataset. The confusion matrix also provides a clearer understanding of the model's fraud-detection performance.

TEST ROC-AUC: 0.973715212379268

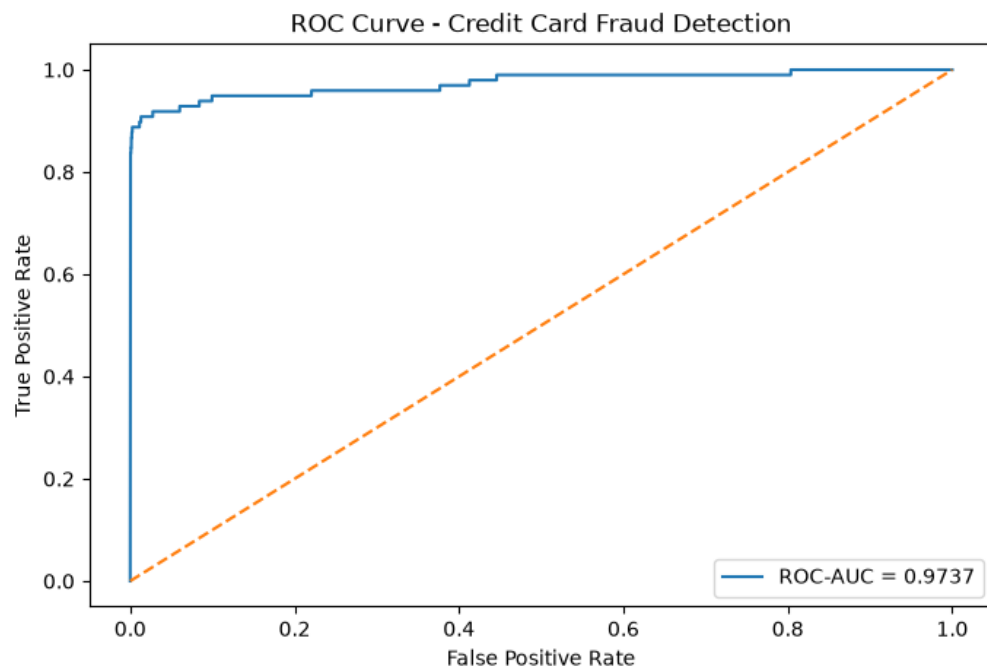


Figure 6.6.1 ROC Curve – Credit Card Fraud Detection

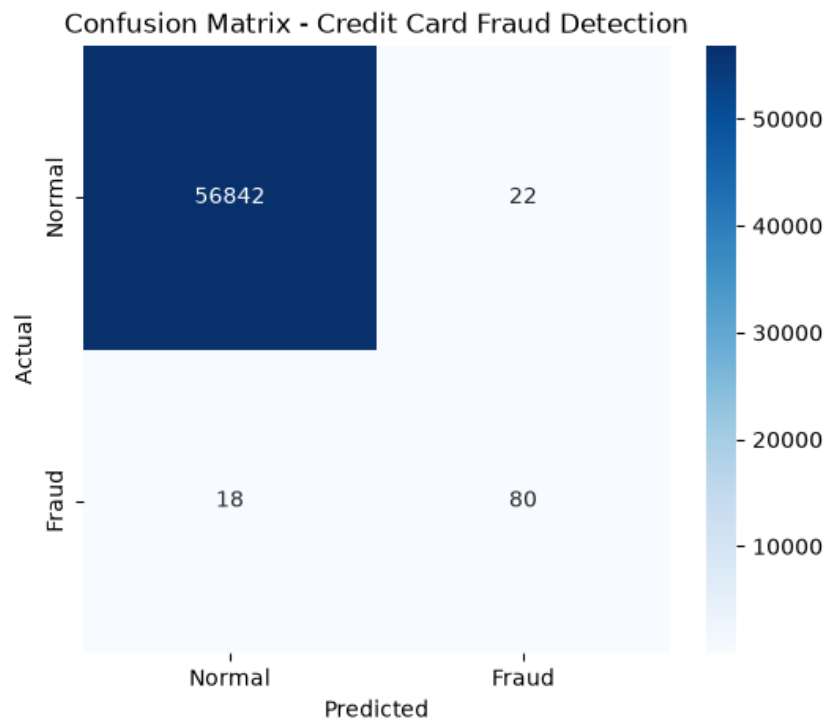


Figure 6.6.2 Confusion matrix Graph

7. Interpretation of Model Results

The Logistic Regression model was interpreted using the magnitude and direction of its feature coefficients. The absolute value of each coefficient was used to determine the relative importance of the corresponding feature in distinguishing fraudulent transactions from legitimate transactions. The analysis identified Amount as the most influential predictor, followed by V1, V14, V4, V10, V12, V5, V2, V17, and V20.

The Amount feature had the highest importance with a coefficient of 2.311589, indicating that transaction amount had a strong influence on the model's fraud prediction. V1 and V4 also had positive coefficients, while V14, V10, V12, V17, and V20 had negative coefficients. In Logistic Regression, the coefficient sign indicates the direction of the feature's association with the predicted class, while the coefficient magnitude indicates its relative influence on the model's decision.

These results show that fraud detection is influenced by a combination of transaction amount and the anonymized transaction features. Since the V1–V28 variables are anonymized features in the dataset, their individual real-world meanings cannot be directly interpreted. However, their coefficients provide useful information about which variables contribute most strongly to the model's fraud classification.

Rank	Feature	Coefficient	Importance
1	Amount	2.311589	2.311589
2	V1	1.652232	1.652232
3	V14	-1.440515	1.440515
4	V4	1.279250	1.279250
5	V10	-1.265119	1.265119
6	V12	-1.176368	1.176368
7	V5	1.120480	1.120480
8	V2	0.994081	0.994081
9	V17	-0.953312	0.953312
10	V20	-0.940625	0.940625

Table 7.1 Top 10 predictors of fraud

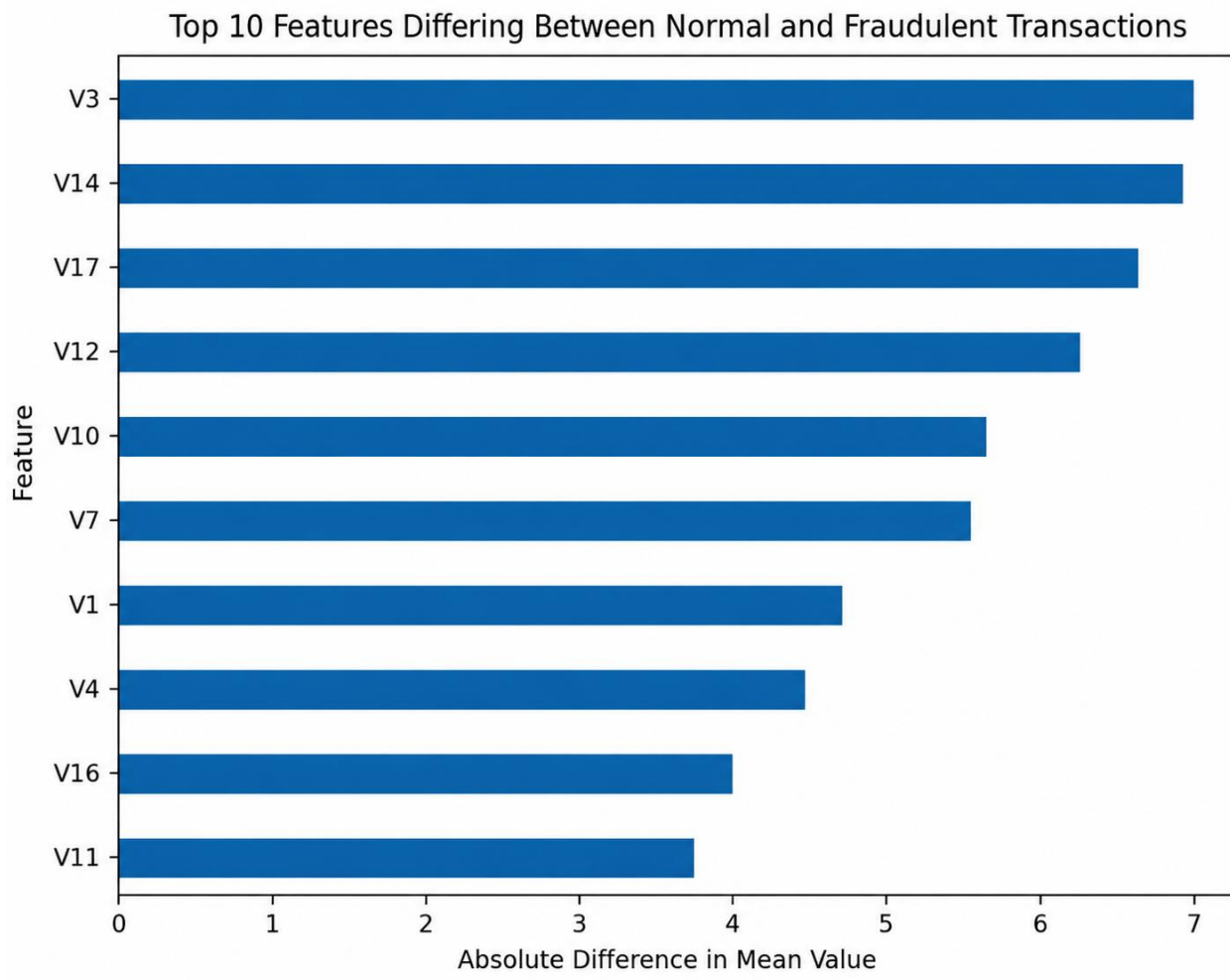


Figure 7.1 Features Differing Between Normal and Fraudulent Transactions

7.1 Top Fraud Predictors

The model identified several features that had a strong influence on distinguishing fraudulent transactions from normal transactions. The most important predictors included Amount, V1, V14, V4, V10, V12, V5, V2, V17, and V20. These features showed the largest absolute differences or coefficients and therefore contributed significantly to the model's fraud-detection decisions.

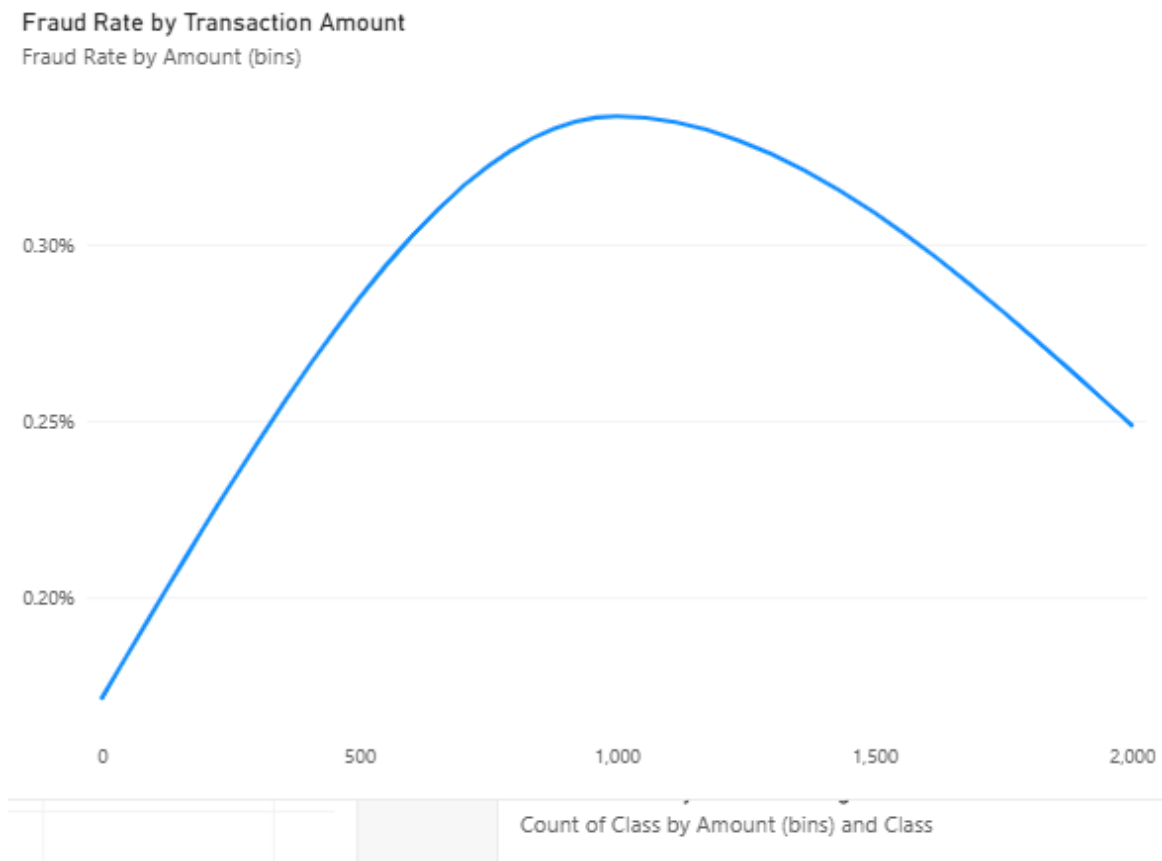


Figure 7.1.1 Top Fraud Predictors of Transaction Amount

8. Power BI Dashboard

The Power BI dashboard provides an interactive visual summary of the fraud detection analysis. It presents key transaction statistics, class distribution, model performance, and important fraud-related patterns, helping users understand the results and derive useful business insights.

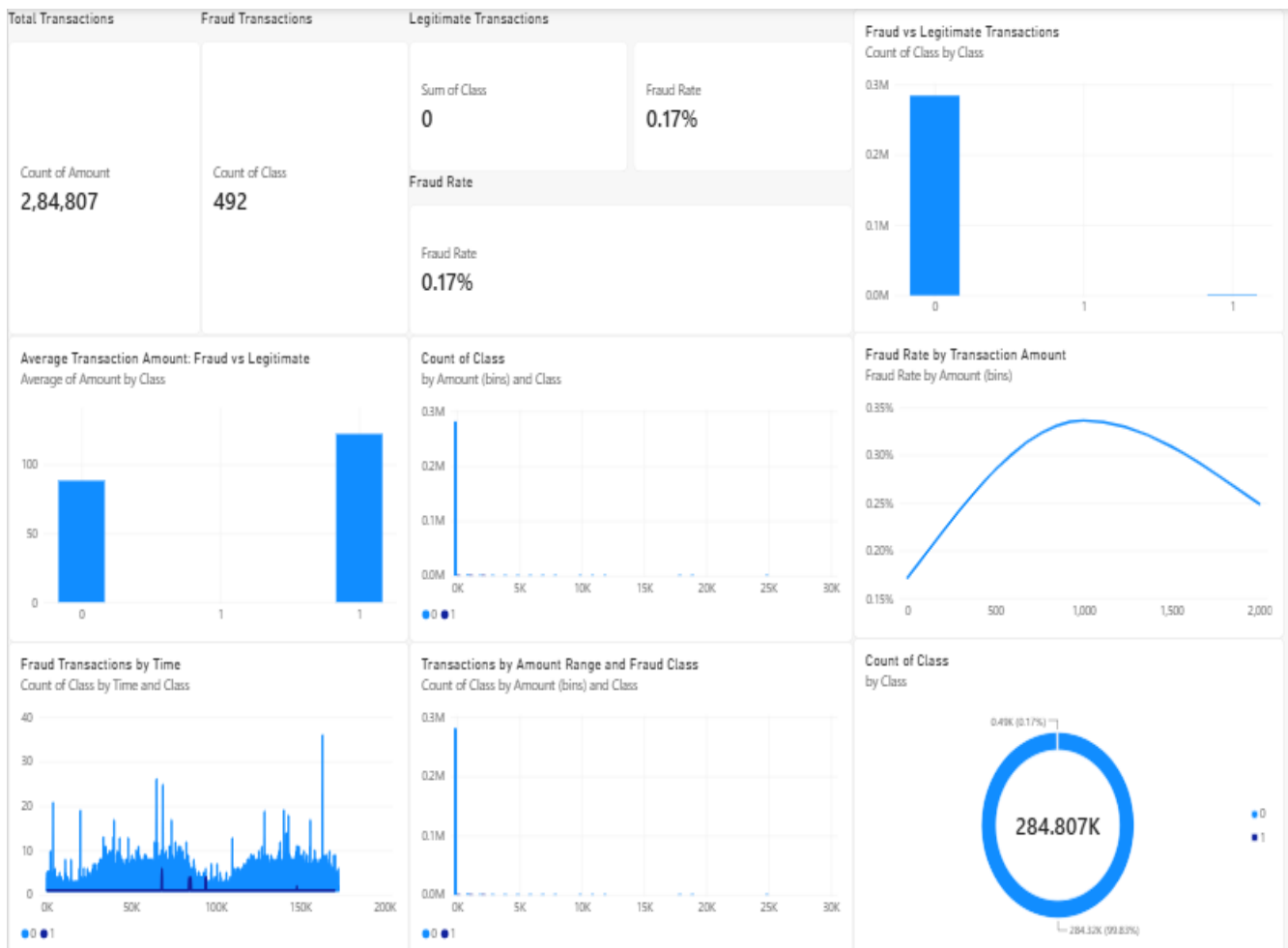


Figure 8.1 Power BI Dashboard

9. Fraud Detection Interface

A user-friendly fraud detection interface was developed using Gradio to demonstrate the practical application of the trained Logistic Regression model. The interface allows users to enter the transaction features, including Time, V1–V28, and Amount. The trained model processes the input and predicts whether the transaction is legitimate or fraudulent, along with the corresponding fraud probability. This interface provides a simple way to demonstrate how the developed fraud detection model can be used for individual transaction prediction.

The screenshot displays a web-based interface for fraud detection. It features a grid of 29 input fields for transaction features: Time, V1 through V28, and Amount. Each field contains a numerical value. At the bottom, there are 'Clear' and 'Submit' buttons. Below the 'Submit' button, a 'Prediction Result' box shows the output: 'NORMAL TRANSACTION' and 'Fraud Probability: 20.53%'. A 'Flag' button is located at the very bottom.

Feature	Value
Time	0.0
V1	-1.359807134
V2	-0.072781173
V3	2.536346738
V4	1.378155224
V5	-0.33832077
V6	0.462387778
V7	0.239598554
V8	0.098697901
V9	0.36378697
V10	0.090794172
V11	-0.551599533
V12	-0.617800856
V13	-0.991389847
V14	-0.311169354
V15	1.468176972
V16	-0.470400525
V17	0.207971242
V18	0.02579058
V19	0.40399296
V20	0.251412098
V21	-0.018306778
V22	0.277837576
V23	-0.11047391
V24	0.066928075
V25	0.128539358
V26	-0.189114844
V27	0.133558377
V28	-0.021053053
Amount	0.0

Prediction Result

NORMAL TRANSACTION
Fraud Probability: 20.53%

Flag

Figure 9.1 Normal Transaction Prediction Using the Fraud Detection Interface

Time 406.0	V9 -2.770089277	V19 0.416955705
V1 -2.312226542	V10 -2.772272145	V20 0.126910559
V2 1.951992011	V11 3.202033207	V21 0.517232371
V3 -1.609850732	V12 -2.899907388	V22 -0.035049369
V4 3.997905588	V13 -0.595221881	V23 -0.465211076
V5 -0.522187865	V14 -4.289253782	V24 0.320198199
V6 -1.426545319	V15 0.38972412	V25 0.044519167
V7 -2.537387306	V16 -1.14074718	V26 0.177839798
V8 1.391657248	V17 -2.830055675	V27 0.261145003
	V18 -0.016822468	V28 -0.143275875

Amount

Clear
Submit

Prediction Result

FRAUDULENT TRANSACTION
Fraud Probability: 100.00%

Flag

Figure 9.2 Fraudulent Transaction Prediction Using the Fraud Detection Interface

10. Conclusion

This project developed a machine learning-based credit card fraud detection system to identify potentially fraudulent transactions. The analysis showed a severe imbalance between legitimate and fraudulent transactions, making appropriate evaluation metrics and class balancing important. The developed model achieved a ROC-AUC score of approximately 0.9737, demonstrating strong capability in distinguishing fraudulent transactions from legitimate ones. The confusion matrix and other performance metrics further helped evaluate the model's effectiveness. The identification of important fraud predictors also provided useful insights into the factors contributing to fraudulent transaction detection.

Conclusion

The credit card fraud detection model was successfully developed and evaluated using the credit card transaction dataset. Due to the highly imbalanced nature of the dataset, Precision, Recall, F1-score, and ROC-AUC were considered important evaluation metrics along with accuracy.

The Logistic Regression model achieved a ROC-AUC score of 0.9737, demonstrating a strong ability to distinguish between legitimate and fraudulent transactions. Using the validation-selected classification threshold, the model achieved a fraud precision of 0.78, fraud recall of 0.82, and fraud F1-score of 0.80 on the unseen test data.

The confusion matrix showed that the model correctly identified 80 fraudulent transactions, while 18 fraudulent transactions were classified as legitimate. Only 22 legitimate transactions were incorrectly classified as fraudulent.

The model coefficients were also analyzed to identify the most influential fraud predictors. Amount, V1, V14, V4, V10, V12, V5, V2, V17, and V20 were identified as the top predictors.

Overall, the developed model demonstrated strong fraud detection performance while maintaining a low number of false positives. The trained model, scaler, and selected classification threshold were saved for future use and deployment.

Figure 10.1 Conclusion

11. Future Scope

The fraud detection system can be further improved by evaluating additional machine learning and deep learning algorithms and comparing their performance. Real-time transaction monitoring could also be incorporated to detect suspicious transactions immediately. Future work could include advanced anomaly-detection techniques, continuous model retraining with new transaction data, threshold optimization, and deployment of the model as a production-level fraud detection application. Integration with interactive dashboards and user-friendly interfaces can further support practical decision-making.

12. References

Kaggle — Credit Card Fraud Detection Dataset.

Scikit-learn — Machine Learning and Model Evaluation Documentation.

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Matplotlib — Python Visualization Library.

Microsoft Power BI Documentation.

Gradio Documentation.